Mahrus Chowdhury

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Education

University of Maryland, College Park — College Park, MD
B.S. in Computer Engineering; ACES Honors College (Cybersecurity) — *Expected May 2028*

Experience

Northrop Grumman — *Incoming Embedded Software Engineering Intern* — *Baltimore, MD | Summer 2027*

Scope Infotech — Remote
AI Software Engineering Intern — *Jul. 2026 – Present*

- Built a Delta Lake medallion lakehouse (PySpark) governing 52 tables with access control for federal-contract demos; added a RAG/semantic search layer (sentence-transformers, FAISS) so non-technical staff can query governed data in plain English.
- Flagged 12/12 planted fraud providers in Medicare claims (K-Means, PCA); cut security alert triage 44% (1,583 → 880) with a CI-gated vulnerability dashboard mapped to CWE, OWASP, and NIST 800-53, demoed to the CISO.

University of Maryland — Intelligent Imaging and Image Processing (I3P) Lab — College Park, MD
Undergraduate Researcher, Dr. Yu Gan — *Jun. 2026 – Present*

- Reproduced nine segmentation benchmarks on ∼5,500 medical images (PyTorch, MMSegmentation, 3× A6000); found two codebases aggregate Dice/IoU differently, fixing the evaluation protocol for a paper in preparation for IEEE.
- Cut preprocessing for 450+ scans from 18 h to 4 h (∼78%), unblocking two models stalled by multi-GPU conflicts.

ThinkEx — Open-Source AI Study Platform (UMD Mokhtarzada Hatchery–funded) — College Park, MD
Software Engineering Intern — *Aug. 2026 – Present*

- Shipped the Sonnet 5.5 upgrade and AI-graded short-answer quizzes; fixed a PostgreSQL race where an in-progress grade could overwrite a quiz reset; synced collaborator edits with Yjs CRDTs. All PRs merged through review and CI.

University of Maryland School of Medicine — Fertig Lab — Baltimore, MD
Researcher — *Nov. 2026 – Present*

- Benchmarking open-weight LLMs (Gemma 4, MedGemma) on spatial transcriptomics reports: whether interpretations preserve quantitative evidence, scored by LLM judges against expert ratings, with runtime/memory profiled on laptop-class hardware.

Team PTERODACTYL — High-Altitude Balloon Payload Program — College Park, MD
Electronics Lead — *2026 – Present*

- Lead the electronics redesign of a near-space balloon payload: migrated a legacy four-layer Fusion/Eagle sensor board to native KiCad and audited it (11 DRC, 33 ERC, 52 parity warnings triaged).
- Found real power-path faults (unprotected regulator OR-ing, USB/battery back-feed, wrong battery-sense rail); drafted the v4 Teensy 4.1 design with BNO055, MS8607, GPS, PTC fuse, reverse-polarity MOSFET, and buck-boost rail. ERC clean; prototype released for fabrication Oct 2026.

A. James Clark School of Engineering — Cyber-Physical Systems Engineering — Shady Grove, MD
Engineering Intern — *Jul. 2025 – Aug. 2025*

- Co-led a 5-person team building Jarvis Sort, an autonomous 3-axis gantry sorting objects in real time at 96.4% accuracy via a CLIP vision pipeline; demoed live to Maryland's Lt. Governor and ∼300 attendees.
- Owned motion-control hardware (TMC2209 over UART, StallGuard homing, 24 V distribution, ESP32) and cut pick-and-place cycles 72% (2.4 s → 0.68 s, 99.1% completion over 1,200 cycles) by bridging vision to embedded control over MQTT.

Projects

Autonomous AI Robotic Air Hockey Table | *Rust, Python, OpenCV, ATmega328P, Control Theory* — Sep. 2024 – May 2025

- Built a 2-axis stepper gantry on bare-metal Rust; improved paddle accuracy ∼25% by pairing a real-time OpenCV puck tracker with closed-loop control. Featured in a UMD ECE article and documentary; presented to Montgomery County officials.

Adapt — Co-founder, Product & Growth | *Open-source chat platform; Rust, PostgreSQL, Redis, SolidJS* — 2022 – 2024

- Co-founded a privacy-focused, open-source (AGPL) alternative to Discord on a Rust microservice backend; drove product and architecture decisions, user research with gaming communities, and growth to ∼1,000 users at peak.

Leadership & Earlier Experience

Claude Builder Club at UMD — College Park, MD
Events & Programming Coordinator — *2026 – Present*

- Bridge Anthropic and UMD students; lead technical workshops for hackathon participants (e.g., EDA with Claude), last one ∼200 attendees.

FTC Team 19818 Jade Innovations — Potomac, MD
Co-captain, Software & Outreach — *2021 – 2025*

- Wrote robot control software in Kotlin: PID with feedforward and gain scheduling, heading-lock, an FSM subsystem manager, inverse kinematics, three-wheel odometry; two FTC World Championship appearances, 2024 Innovate Award.
- Secured sponsorships from Bambu Lab, Fabworks, GenScript, UnitedHealthcare, Hughes, and UMD CPS Engineering; 900+ outreach hours, including a four-week Python course at Germantown Library and mentoring three rookie FLL teams.

Technical Skills

Software & Data: Python, Rust, C/C++, Java, Kotlin, SQL, TypeScript, PySpark, Delta Lake, PyTorch, OpenCV, FAISS/RAG, PostgreSQL, Docker, Linux, Git
Embedded & Hardware: STM32, ESP32, AVR, KiCad (multi-layer PCB), I^2C/SPI/UART, MQTT, LoRa, PID control, oscilloscope, SMD reflow